

Technical Exercise — Sales KPI Dashboard

Full-stack take-home · React + FastAPI + Postgres · AI-assisted with Claude Code

Welcome, and thanks for taking the time. This exercise is about **working effectively with an AI coding assistant (Claude Code)** while genuinely **understanding** the code you ship — not just producing it.

You'll work in the provided repository — a small but real full-stack app (React + FastAPI + Postgres). **Clone it and follow the setup in its README:**

```
git clone https://github.com/WFH-Holding-Group/technical_test_ops_app.git
```

Get it running first and click around so you understand what it does.

There are **four parts**. Use Claude Code freely throughout. We care as much about *how you think and explain* as about the final diff.

Suggested time: ~2–3 hours for Parts A–C, then ~20–30 minutes to prepare Part D. Commit your work as you go so we can follow your progress.

Questions? If you get stuck or need any clarification, reach out to Jeremy at jeremy@findyourtribe.co — we're happy to help.

Part A — Fix the bugs 🐛

This dashboard has **several correctness bugs**: some of the numbers it reports are wrong. We won't tell you where they are or how many — finding them is part of the exercise.

- Explore the app, compare what it shows against what the data *should* produce, and track down the causes.
- Fix them cleanly.
- For each bug, be ready to explain in Part D: **what was wrong, why it produced the wrong result, and how your fix addresses it.**

Tip: the sample data is deterministic, so you can reason about expected totals by hand or with a quick script.

Part B — Build an enhancement ✨

Add a "Pace to Target" panel that forecasts whether the selected month will hit its sales target, based on the pace so far.

For the selected month, show:

Metric	Meaning
Month-to-date revenue	net revenue from completed sales so far this month
Days elapsed / days in month	how far into the month we are
Projected month-end revenue	run-rate forecast = $\text{MTD} \div \text{days elapsed} \times \text{days in month}$
Target	the month's target
Pace vs target	projected vs target (RM and %), with an on-track / at-risk indicator

Handle these cases correctly (this is the point of the exercise):

- A **past** (already finished) month → there is no remainder to project; projected should equal the actual month total.
- A **future** month → nothing has happened yet (0 days elapsed) → must not divide by zero; show a sensible "N/A".
- Months have **28/29/30/31** days (February in a leap year is 29).
- Use the **correct calendar date for "today"** (this project's timezone is Malaysia, UTC+8).
- No **NaN** / **Infinity** when a target is missing or 0 days have elapsed.

You may add a backend endpoint and a frontend panel, in whatever shape you think is clean. Be ready to explain your edge-case handling.

Part C — Innovation 💡

Add **one improvement of your own choosing** — something you think makes the product genuinely better. Scope it so you can finish and explain it.

Some directions (you are **not** limited to these): CSV export, a revenue trend sparkline, a refund-rate metric, a "top products" ranking, inline editing, keyboard shortcuts, dark mode, better empty/loading states, tests.

Be ready to explain *why* you picked it and what it adds.

Part D — Present it 🎤

Prepare a **short slide deck** (≈5–8 slides) and walk us through it. Use Claude to help you design/generate the slides quickly — that's encouraged.

Cover:

- **Bugs** — what you found, why each was wrong, how you fixed it.
- **Enhancement** — how your Pace-to-Target feature works and how you handled the tricky cases.
- **Innovation** — what you built and why.

Also touch briefly on **how you used Claude Code** — where it helped, and where you had to steer or correct it.

Submitting your work

When you're finished, share your work as a **GitHub repository** and reply to our email with the link.

- **Commit as you go** in small, meaningful steps — we like to see how your work progressed, not just the final state (please don't squash or force-push over your history).
- Push everything to **your own public GitHub repo** (you won't have push access to the original).
- Include your **Part D slides** in the repo (or a link to them) alongside the code.
- **Reply to our email with the GitHub repository link** once you're done.

What we're evaluating

- **Correctness** — are the numbers right, and are the edge cases handled?
- **Understanding** — can you clearly explain *why*, not just *what*?
- **Code quality** — clean, readable, consistent with the existing code.
- **Effective AI use** — using Claude Code to move fast *and* stay in control.
- **Communication** — a clear, confident walkthrough.

Good luck — have fun with it!